

Semantic similarity & clustering (document embeddings)

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From words to whole documents

The "Word embeddings with BERT" and "Word2Vec with Gensim" features embed individual words — each word becomes a vector, and words used in similar contexts land near each other. This feature works one level up: it embeds entire documents. Each text in your corpus is turned into a single vector that captures its overall meaning, so you can measure how semantically close two documents are and group documents that talk about the same things — even when they share few of the same words.

How it works: SBERT (Sentence-BERT)

Plain BERT produces one vector per word/token, which is awkward for comparing whole documents. SBERT (Sentence-BERT, the sentence-transformers library; Reimers & Gurevych 2019) is a variant fine-tuned so that a single fixed-length vector represents a whole passage, and the cosine similarity between two such vectors reflects how similar their meanings are. NLP Suite uses the compact, fast all-MiniLM-L6-v2 model. It is bundled with the NLP Suite; the model file itself (~80 MB) is downloaded automatically the first time you run the feature, so the first run needs an internet connection.

What the feature computes

Given a corpus (a directory of at least two .txt files), it produces:

A document × document similarity matrix (NLP_semantic_similarity_matrix.csv) — the cosine similarity of every pair of documents (1.0 = identical meaning, 0 = unrelated).

An interactive heatmap (NLP_semantic_similarity_heatmap.html) — the same matrix visualized; bright cells are the most similar document pairs.

Document clusters by meaning (NLP_semantic_clusters.csv) — the documents grouped with K-means, where the number of clusters is chosen automatically (best silhouette score). Documents in the same cluster are thematically alike.

When to use it

Find near-duplicate or closely related documents in a large corpus.

Group a corpus into themes/topics without predefined categories (a semantic complement to topic modelling).

Spot outlier documents that don't resemble the rest.

Compare the meaning of texts that use different vocabulary to say similar things (where word-overlap measures such as n-grams fail).

How this differs from other semantic options

Word embeddings (BERT) / Word2Vec — vectors for words, not documents.

WordNet/VerbNet/FrameNet aggregation & WSD — map words to lexical categories/senses from a dictionary; here nothing is mapped to a dictionary — similarity comes straight from the neural embedding.

Topic modelling (LDA) — discovers topics as word distributions; this instead measures document-to-document distance directly and clusters on it.

How to run it in NLP Suite

Semantic Analysis GUI → More semantic analyses dropdown → Semantic similarity & clustering (document embeddings). Select your corpus (a directory of txt files) in the INPUT/OUTPUT configuration and press RUN.

References

Reimers, Nils, and Iryna Gurevych. 2019. "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks." Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP), 3982–3992. Association for Computational Linguistics.